TacEnvelope: A Plug-in Framework for Utilizing Tactile Envelope Feedback in RL Post-Training of VLA Policies

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***Abstract*— Reinforcement learning (RL) post-training is an effective approach for refining task-adapted Vision-Language-Action (VLA) policies, but task success or failure signals provide limited information about physical interaction quality in contact-rich manipulation. Tactile signals provide rich contact information, yet these signals alone do not specify which contact behaviors should be encouraged or penalized during policy refinement. To address this challenge, we present TacEnvelope, a plug-in framework that incorporates adaptive tactile feedback into online RL post-training of task-adapted VLA policies, while requiring only limited additional tactile data and modest computational overhead. Its tactile envelope expert learns to predict action-conditioned short-horizon contact load from successful demonstrations and constructs calibrated force envelopes that adapt to the ongoing interaction. During execution, the framework uses a dual-rate inference scheme to evaluate contact over shorter local windows within longer action chunks. Deviations between the observed contact load and the predicted envelope are converted into tactile feedback as reward for online RL post-training. We validate TacEnvelope on real-world USB insertion, ink dispensing, and force-sensitive pastry grasping. Compared with the respective RLT and DSRL baselines, TacEnvelope improves insertion and dispensing success rates by 5–12.5 percentage points, reduces USB completion time over successful trials by 35–37% and dispensing volume error by 33–60%, and increases pastry preservation scores by 58–100%.**

I. Introduction

Vision-Language-Action (VLA) models have demonstrated strong capabilities in robotic manipulation [1], [2]. Reinforcement learning (RL) post-training can further refine these policies through interaction with the environment. However, when refining these policies through RL, task success or failure signals provide limited information about physical interaction quality in contact-rich manipulation [3], [4], [5]. Two trajectories may both complete a manipulation task successfully, while one proceeds smoothly and the other involves excessive force or unstable contact [3]. Likewise, failed trajectories may exhibit substantially different contact behaviors, which cannot be differentiated using failure signals alone. Therefore, policy refinement in contact-rich manipulation requires feedback beyond task success to characterize the contact feedback.

Tactile sensing provides such information by directly capturing contact during physical interaction [4], [5]. Existing tactile-aware policies use this information for action generation or refinement [4], [5], [6], [7], but incorporating tactile inputs does not itself provide an explicit reward for desirable contact behavior. Tactile-derived rewards offer a direct way to supervise interaction quality, with prior work encouraging contact or penalizing excessive force through predefined objectives [8], [9]. However, these objectives do not explicitly capture how the expected contact load varies with the current state and intended action, limiting their ability to distinguish appropriate contact from undesirable force deviations.

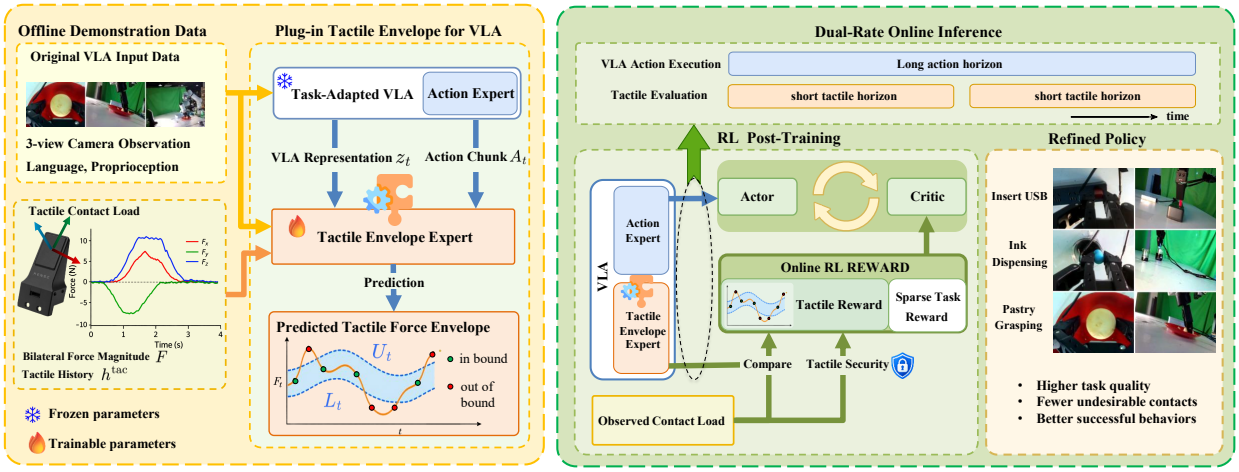
We therefore introduce TacEnvelope, a plug-in framework that incorporates adaptive tactile reward feedback into online RL post-training without changing the VLA backbone architecture. The module conditions on VLA latent representations, action chunks, and recent tactile history to predict the expected contact load from successful demonstrations and construct a calibrated force envelope for the current interaction. Deviations of the observed contact load from the predicted force envelope provide tactile feedback on the realized physical interaction: contact load below the expected range can indicate weak or lost contact, whereas contact load above the expected range can indicate collision, jamming, or excessive compression.

In contact-rich manipulation, physical interactions often require more frequent tactile assessment than VLA action inference. TacEnvelope therefore adopts a dual-rate inference scheme to assess contact changes within VLA action chunks. While the VLA retains its original execution horizon, the framework evaluates physical interaction over short windows. Each predicted envelope is compared with the observed contact load over the same window. This enables multiple tactile evaluations within each action chunk.

Our contributions are threefold:

- We propose TacEnvelope, a plug-in framework that evaluates physical interaction quality and provides dense, adaptive tactile feedback as reward for online RL post-training of task-adapted VLA policies.
- At the core of TacEnvelope is a tactile force envelope expert that predicts calibrated short-horizon force envelopes. We further introduce a dual-rate inference scheme that enables repeated tactile evaluation within each VLA action chunk while preserving the original VLA execution horizon.
- We validate TacEnvelope on three real-world contact-rich manipulation tasks. When augmented with TacEnvelope, both RL Token (RLT) [10] and DSRL [11] outperform their respective baselines in task success and physical interaction quality.

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(a)Offline-training stage: Tactile Envelope Expert Training (b)Online-training Stage: RL Post-training with tactile feedback

Fig. 1. Overview of TacEnvelope, a plug-in framework for tactile envelope feedback in online RL post-training of VLA policies. (a) Offline training: a tactile envelope expert, attached to the task-adapted VLA as an additional prediction head, learns action-conditioned short-horizon contact load distributions from successful demonstrations while the VLA backbone remains frozen. Its predictions are calibrated to construct tactile force envelopes. (b) Online RL post-training: deviations of observed contact load from the corresponding predicted envelopes generate dense tactile reward that complement sparse task reward. A dual-rate inference scheme enables multiple tactile evaluations within each VLA action chunk while preserving the original action execution horizon. Each envelope is compared with the contact load observed over the same local window.

II. RELATED WORK

A. RL-Based Policy Post-Training

Recent work has explored RL-based post-training for pretrained robot policies. iRe-VLA [12] combines online RL with supervised learning for VLA post-training, while VLA-RL [13] introduces process level supervision for online policy optimization. RL Token [10] performs lightweight actor–critic refinement using a compact VLA representation, while DSRL [11] performs RL in the latent noise space of diffusion-based policies. These studies explore different mechanisms for RL-based post-training, including process level supervision, lightweight actor–critic learning, and latent-space optimization. For RL post-training in contact-rich manipulation, an additional challenge is to provide informative feedback about the quality of physical interaction during policy optimization. To address this challenge, TacEnvelope learns calibrated force envelopes and constructs adaptive tactile rewards for online RL post-training.

B. Tactile-Aware VLA Policies

Recent tactile-aware VLA methods incorporate tactile signals directly into VLA policies to improve contact-rich manipulation. VTLA [5] jointly models vision, tactile signals, language, and actions for contact-rich manipulation. ForceVLA [14] treats tactile signals as a first-class modality and introduces a tactile-aware mixture-of-experts module that fuses real time tactile signals with visual-language representations during action generation. ForceVLA2 [15] further introduces explicit force-aware concepts and hybrid force position control for contact-rich manipulation. VLA-Touch [6] preserves the base VLA while introducing tactile-language feedback for high level reasoning and a tactile-conditioned interpolant policy that refines VLA-generated actions.

These methods demonstrate the value of tactile information for manipulation, but directly integrating tactile signals into the policy often requires additional data collection and training. In contrast, TacEnvelope keeps the task-adapted VLA unchanged and trains only a lightweight plug-in envelope expert, then incorporates tactile feedback into RL post-training with limited tactile data and additional training overhead.

C. Tactile Feedback in RL-Based Methods

Tactile feedback has also been exploited as a supervisory signal for contact-rich manipulation. Tac2Motion [16] incorporates tactile observations and hand-designed tactile reward shaping into RL for dexterous manipulation. Such manually specified rewards require redesign or retuning when transferred to different interaction tasks. HapticVLA [8] uses tactile-derived rewards during offline policy learning and subsequently distills tactile-aware behavior into a VLA that does not require tactile sensing at inference time, and thus does not provide online tactile evaluation of the realized interaction. TORL-VLA [7] uses tactile-derived wrench history to predict reference actions and future wrench sequences, which guide a lightweight online RL module in refining the actions. However, undesirable contact does not receive an explicit tactile penalty and may therefore persist even when the task is successfully completed.

In contrast, TacEnvelope learns a force envelope from successful demonstrations and explicitly converts tactile signals into tactile rewards. This provides physical interaction quality supervision for online RL post-training.

III. METHOD

As illustrated in Fig. 1, TacEnvelope combines offline tactile envelope learning with online RL post-training. Section III-A introduces the plug-in tactile envelope expert and its calibration procedure (Fig. 1(a)). Section III-B presents the dual-rate inference scheme, and Section III-C describes how tactile feedback is incorporated into online RL post-training (Fig. 1(b)).

A. Plug-in Tactile Envelope Expert

The proposed tactile envelope expert is designed to estimate the expected short-horizon contact load associated with the task-adapted VLA action chunk, while remaining external to the VLA policy. For each tactile side $s \in \{\ell, r\}$, where ℓ and r denote the left and right tactile sensors shown in Fig. 4, respectively, $F_{t,s}$ denotes the resultant contact force magnitude reported by that sensor at time t.

Let H_T denote the number of control steps in the tactile prediction horizon and Δt the control period. The corresponding window duration is therefore $T_w = H_T \Delta t$. We construct a contact-load target for each tactile side over this future window. Let τ_t denote the physical timestamp corresponding to control step t. Since tactile measurements arrive asynchronously, samples are assigned to the window according to their timestamps without resampling or interpolation:

$$\mathcal{W}_{t,s} = \{F_{i,s} \mid \tau_t \le \tau_{i,s} < \tau_t + T_w, \mathrm{valid}(i,s)\}, \quad (1)$$

where i indexes tactile samples, $\tau_{i,s}$ is the sampling timestamp on side s, and $F_{i,s}$ is the preprocessed force magnitude. Here, $\mathrm{valid}(i,s)$ indicates that the sample corresponds to a non-duplicate sensor frame, so repeated frames are excluded from the tactile window, as illustrated in Fig. 2(b). The supervision target is

$$y_{t,s} = Top2Mean(\mathcal{W}_{t,s}), \quad (2)$$

where $Top2Mean$ averages the two largest valid force magnitudes within the window, separately for the left and right sides.

The plug-in envelope expert predicts a Gaussian distribution for each tactile side:

$$(\mu_{t,s}, \sigma_{t,s}) = f_\theta\left(z_t, q_t, A_t^{H_T}, h_t^{\mathrm{tac}}\right), \quad (3)$$

where z_t denotes the VLA latent representation, q_t the robot proprioceptive state, $A_t^{H_T}$ the action slice over the tactile prediction horizon H_T, and h_t^{tac} the recent tactile history. The outputs $\mu_{t,s}$ and $\sigma_{t,s}$ represent the predicted mean contact load and its input-dependent uncertainty scale, respectively.

The corresponding supervision target is modeled as

$$y_{t,s} \mid z_t, q_t, A_t^{H_T}, h_t^{\mathrm{tac}} \sim \mathcal{N}\left(\mu_{t,s}, \sigma_{t,s}^2\right). \quad (4)$$

The envelope expert is optimized using a Gaussian negative log-likelihood (NLL) together with a SmoothL1 regression term on the predicted mean. The Gaussian NLL $\mathcal{L}_{\mathrm{NLL},s}$ is defined as

$$\mathcal{L}_{\mathrm{NLL},s} = \frac{1}{2}\left[(y_{t,s} - \mu_{t,s})^2 e^{-\ell_{t,s}} + \ell_{t,s}\right], \quad (5)$$

where $\ell_{t,s} = \log \sigma_{t,s}^2$. The resulting envelope-training objective is

$$\mathcal{L}_{\mathrm{env},s} = \mathcal{L}_{\mathrm{NLL},s} + \lambda_{\mathrm{reg}} SmoothL1\left(\mu_{t,s}, y_{t,s}\right), \quad (6)$$

where λ_{reg} controls the regression term, and $SmoothL1$ denotes the smooth L_1 regression loss [17]. The left- and right-side losses are computed independently over valid samples and averaged to obtain the final training objective.

For bilateral envelope construction, we first aggregate the predicted means and tactile targets into bilateral averages,

$$y_t^{\mathrm{avg}} = \frac{y_{t,\ell} + y_{t,r}}{2}, \qquad \mu_t^{\mathrm{avg}} = \frac{\mu_{t,\ell} + \mu_{t,r}}{2}. \quad (7)$$

We further define an input-dependent bilateral uncertainty scale as

$$u_t = \frac{\sigma_{t,\ell} + \sigma_{t,r}}{2}. \quad (8)$$

Here, u_t is used as an empirical uncertainty scale rather than the standard deviation of the bilateral average, avoiding assumptions on the dependence between the two tactile-side prediction errors.

After training, we freeze the tactile expert and use separate demonstration episodes, excluded from training and model selection, to calibrate its predicted uncertainty against measured contact loads. This converts the predicted mean and uncertainty scale into a nominal 90% force envelope. For each calibration window, we compute the dimensionless standardized residual:

$$s_i = \frac{y_i^{\mathrm{avg}} - \mu_i^{\mathrm{avg}}}{\max(u_i, \epsilon)}, \quad (9)$$

where $\epsilon > 0$ is a small constant for numerical stability. The lower and upper calibration coefficients are then obtained from the 5th and 95th percentiles of these standardized residuals:

$$\alpha_{05} = Q_{0.05}(s), \qquad \alpha_{95} = Q_{0.95}(s), \quad (10)$$

where $Q_p(s)$ denotes the empirical p-quantile of the standardized residuals on the calibration set.

The calibrated tactile force envelope is then defined as

$$L_t = \mu_t^{\mathrm{avg}} + \alpha_{05} u_t, \qquad U_t = \mu_t^{\mathrm{avg}} + \alpha_{95} u_t. \quad (11)$$

The resulting interval $[L_t, U_t]$ therefore adapts both its center and width to the current interaction context. Fig. 3(a) illustrates a representative USB insertion rollout, where the calibrated envelope tracks the expected bilateral contact load while deviations from the operating range remain directly observable.

B. Dual-Rate Inference Scheme

VLA policies predict relatively long action chunks to maintain motion continuity, while contact dynamics evolve at a shorter temporal scale. We therefore adopt the dual-rate inference scheme.

To accommodate VLA inference latency, we employ an RTC-based asynchronous execution mechanism [18], which overlaps the generation of the next action chunk with execution of the current one. Let H_a denote the VLA action-prediction horizon and H_e the number of control steps executed per execution cycle. To assess contact changes within each cycle, we refresh the tactile envelope over short windows of H_T control steps, as illustrated in Fig. 2(a). The choice of H_e balances motion continuity and replanning responsiveness, while H_T balances timely contact assessment with sufficient valid measurements at the effective tactile sampling rate.

In our online RL implementation, each execution cycle corresponds to one RL transition. For the n-th execution cycle starting at control step t, we divide the H_e executed steps into K tactile windows:

$$H_e = KH_T, \qquad H_T < H_e \leq H_a. \tag{12}$$

The j-th window, $j \in \{1, \ldots, K\}$, starts at control step $t+(j-1)H_T$. Accordingly, its predicted envelope is denoted by

$$[L_{n,j}, U_{n,j}] = [L_{t+(j-1)H_T}, U_{t+(j-1)H_T}]. \tag{13}$$

The same (n, j) indexing is used for window-level tactile costs, which are aggregated into the reward r_n of transition n.

At the start of each tactile window, the envelope expert is queried using the corresponding action slice and updated robot proprioception and tactile history. After the window ends, the observed contact load is aggregated and compared with the predicted envelope to compute tactile feedback. The window-level feedback is aggregated into a single reward for the RL transition, as described in Section III-C.

C. Tactile Feedback for RL Post-Training

During online RL post-training, TacEnvelope converts tactile feedback into an additional reward term. For each tactile evaluation window, we compute the realized bilateral contact load using the same preprocessing and aggregation rule as in offline envelope learning. Given the calibrated envelope $[L_t, U_t]$, deviations below and above the expected contact are defined as

$$d_t^{\text{low}} = \max(0, L_t - y_t^{\text{avg}}), \tag{14}$$

and

$$d_t^{\text{high}} = \max(0, y_t^{\text{avg}} - U_t). \tag{15}$$

The example tactile reward is shown in Fig. 3(b), where deviations from the calibrated envelope produce negative feedback as dense reward for policy optimization.

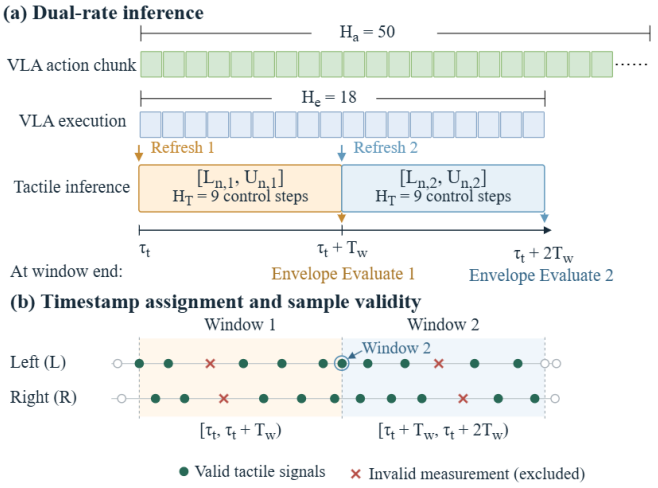


Fig. 2. Dual-rate inference and asynchronous tactile-window construction. (a) A VLA action chunk is executed over a longer horizon H_e, while the tactile envelope is evaluated over shorter windows of length H_T. Each window produces one tactile evaluation without triggering additional VLA replanning. (b) Asynchronous tactile measurements are assigned to half-open temporal windows according to their timestamps; invalid measurements are excluded.

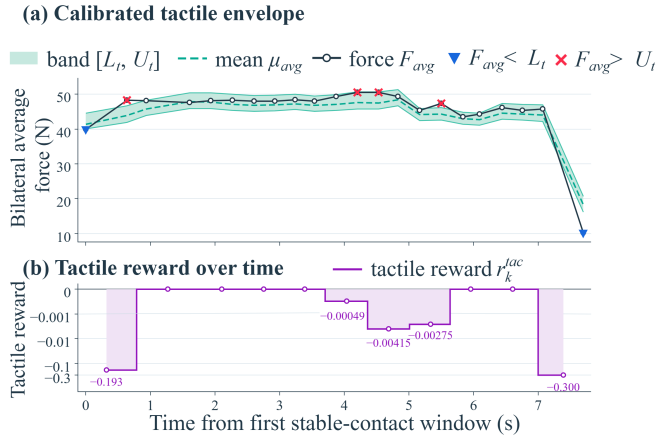


Fig. 3. Illustration of tactile envelope-based contact evaluation on a representative USB insertion rollout. (a) The calibrated tactile force envelope $[L_t, U_t]$ is compared with the observed bilateral contact load y_t^{avg}. (b) The resulting tactile reward over time, where deviations from the calibrated envelope produce negative tactile feedback for policy optimization.

The soft penalty encourages contact loads consistent with successful demonstrations under the current interaction context, penalizing both insufficient and excessive loading. For windows with established contact, we define:

$$c_t^{\text{soft}} = clip\left(\frac{d_t^{\text{low}}}{F_{\text{scale}}}, 0, 1\right)^2 + clip\left(\frac{d_t^{\text{high}}}{F_{\text{scale}}}, 0, 1\right)^2, \tag{16}$$

where $F_{\text{scale}} > 0$ is a force-deviation scale.

For each RL transition, we average the soft costs over its K tactile windows:

$$\bar{c}_n^{\text{soft}} = \frac{1}{K} \sum_{j=1}^{K} c_{n,j}^{\text{soft}}. \tag{17}$$

This preserves the soft penalty scale across different tactile evaluation frequencies.

To capture abrupt unilateral force signals that bilateral averaging may attenuate, we introduce a peak force term:

$$c_t^{\text{peak}} = I\left[\max_{s,i} \widetilde{F}_{i,s}^{\text{raw}} > F_{\text{peak}}\right], \tag{18}$$

where the maximum is taken over valid tactile samples from both sides within $[\tau_t, \tau_t + T_w)$. Here, $\widetilde{F}_{i,s}^{\text{raw}}$ denotes the bias-corrected but unfiltered force magnitude, and F_{peak} denotes the device-specific safety force threshold.

We retain any peak force event within the transition by taking the maximum over its tactile windows:

$$c_n^{\text{peak}} = \max_{j=1,\ldots,K} c_{n,j}^{\text{peak}}. \tag{19}$$

The online RL transition-level tactile penalty combines the averaged soft envelope cost and the peak force event:

$$p_n = \lambda_{\text{soft}} \bar{c}_n^{\text{soft}} + \lambda_{\text{peak}} c_n^{\text{peak}}, \tag{20}$$

where λ_{soft} and λ_{peak} control the contributions of the two terms. The resulting reward is

$$r_n = r_n^{\text{task}} - p_n, \tag{21}$$

where r_n^{task} equals 1 upon task success and 0 otherwise.

Taking RLT as an example, the frozen VLA provides an RL token and a reference action chunk, which condition a lightweight actor to generate executable actions. During robot execution, TacEnvelope evaluates the measured contact load within each tactile window and aggregates the resulting costs into p_n. The shaped reward r_n is then used in RLT's original critic and actor optimization. The VLA and tactile envelope expert remain frozen during online RL. We apply the same reward augmentation to DSRL while retaining its original optimization procedure.

IV. EXPERIMENTS

A. Experimental Setup

Robot platform and tactile sensing. All experiments are conducted on a real world single-arm manipulation platform consisting of a Franka Research 3 (FR3) [19] robot equipped with a Robotiq parallel-jaw gripper, as shown in Fig. 4(a). The VLA receives RGB observations from three views, including two external cameras and one wrist-mounted camera. Two XSense G1-WS visuo-tactile sensors [20] are mounted on the opposing fingertips of the gripper to capture bilateral contact information during manipulation, as shown in Fig. 4(b).

The estimated contact forces are used by TacEnvelope to construct the contact-load signal for tactile evaluation and reward computation, but are not provided as inputs to the VLA policy.

Tasks and evaluation metrics. We evaluate the proposed framework on three real world contact-rich manipulation tasks (Fig. 5), using task-specific evaluation protocols:

- **USB insertion:** The robot inserts a USB connector into a target port. Each method is evaluated over 40 trials. We report the success rate and mean completion time over successful trials, measured from the start of insertion execution to successful insertion.
- **Fixed-volume ink dispensing:** The robot performs 10 consecutive dispensing operations in each evaluation trial, with a target volume of 5 mL for each operation. After each operation, the dispensed volume v_i is measured using a graduated pipette, with $|v_i - 5| \leq 1\,\text{mL}$ counted as success. We compute the accumulated absolute volume error as $\sum_{i=1}^{10} |v_i - 5|$, averaged over four independent trials.
- **Pastry grasping:** The robot repeatedly grasps and transfers two types of delicate pastries while minimizing physical damage. After 20 consecutive grasp-and-transfer cycles, each pastry type is scored independently using a four-level preservation scale: 3–*intact*, 2–*slight crack*, 1–*crack*, and 0–*fracture* (Fig. 6). The two scores are summed to obtain a combined preservation score ranging from 0 to 6, with higher scores indicating better preservation. We report the mean combined score over five independent trials.

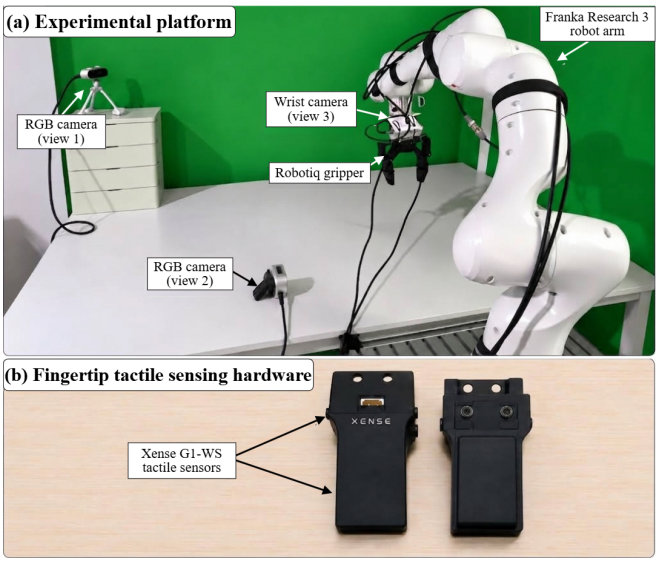


Fig. 4. Real world experimental platform and tactile sensing hardware. (a) Franka Research 3 robot equipped with a Robotiq parallel-jaw gripper and three RGB views, including two external cameras and one wrist-mounted camera. (b) XSense G1-WS visuo-tactile sensors used for bilateral fingertip contact sensing.

Together, these tasks cover diverse contact regimes, including constrained insertion, controlled dispensing, and delicate grasping, all of which require continuous and precise control of physical contact.

Evaluation protocol. We evaluate the proposed tactile-envelope feedback with two online RL post-training backends, RLT [10] and DSRL [11]. For each backend, we compare RL post-training using sparse task rewards alone against RL post-training with additional tactile reward. Performance is evaluated using both task success and task-specific measures of physical interaction quality.

Implementation details. The robot is controlled at 30 Hz.

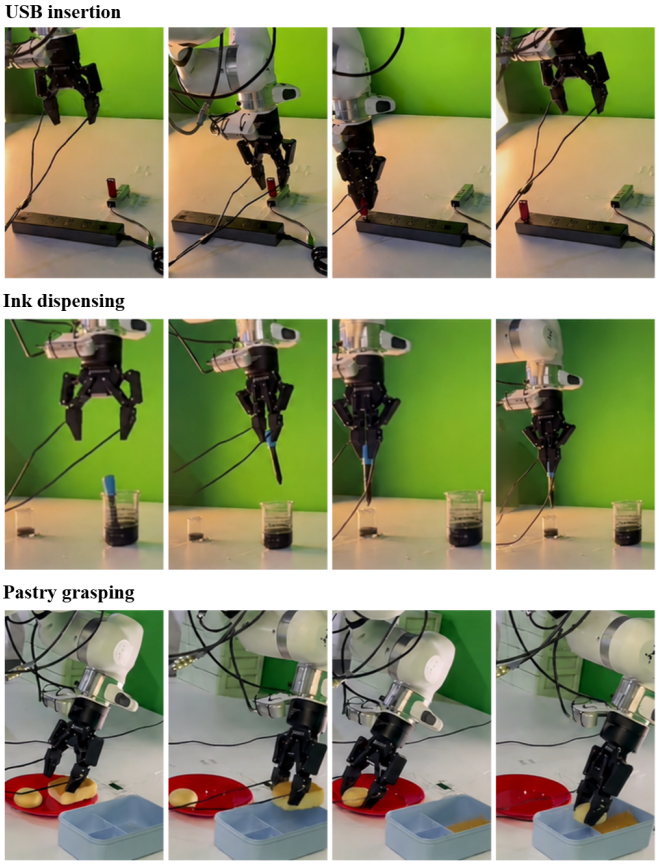


Fig. 5. Real world contact-rich manipulation tasks used for evaluation: USB insertion, ink dispensing, and pastry grasping. The tasks cover constrained insertion, controlled dispensing, and delicate grasping.

Representative pastry preservation scores

Type \ Level	Intact (3)	Slight crack (2)	Crack (1)	Fracture (0)
Cake				
Mung bean cake				

Fig. 6. Examples of the pastry preservation score used for evaluation: 3–*intact*, 2–*slight crack*, 1–*crack*, and 0–*fracture*. Examples are shown for both cake and mung bean cake.

Each VLA inference predicts an action chunk of $H_a = 50$ actions, while each RL transition executes $H_e = 18$ control steps (0.6 s). Tactile feedback is evaluated over windows of $H_T = 9$ control steps (0.3 s), yielding $K = 2$ tactile evaluations per RL transition.

B. Tactile Force Envelope Prediction Evaluation

We first evaluate whether the proposed envelope expert can predict short-horizon contact load and construct calibrated uncertainty-aware force envelopes during physical interaction. Complete episodes are split into training, validation, calibration, and test subsets at the episode level. The predictor is evaluated only on windows in which bilateral contact has been established, while the calibration split is used exclusively to estimate the standardized-residual quantiles for envelope construction. Table I summarizes the prediction performance across the three manipulation tasks.

The proposed model achieves MAEs of 1.32 N, 0.80 N, and 0.60 N for USB insertion, ink dispensing, and pastry grasping, respectively. Relative to the mean contact force in each task, these errors correspond to 2.59%, 3.97%, and 17.26%. The higher relative error for pastry grasping mainly reflects its much lower force scale.

The nominal 90% uncertainty-scaled envelopes achieve coverages of 89.80%, 88.79%, and 91.25% for the three tasks, respectively, remaining close to the target coverage across substantially different contact regimes. The representative USB rollout in Fig. 3(a) further illustrates that both the predicted contact level and envelope width evolve with the ongoing interaction. These results indicate that the proposed expert provides an adaptive tactile reference for short-horizon contact evaluation.

C. Online RL Post-Training with Tactile Feedback

We use the task-adapted $\pi_{0.5}$ as base policy before RL post-training, and instantiate two online RL backends, RLT and DSRL. For each backend, we compare the original methods with the same methods using TacEnvelope.

Table II reports the results for USB insertion and ink dispensing. For USB insertion, TacEnvelope improves both success and execution quality: RLT increases from 72.5% to 85% success while reducing completion time from 19 s to 12 s, and DSRL increases from 87.5% to 95% while reducing completion time from 20 s to 13 s. Similar improvements are observed for ink dispensing, where TacEnvelope increases success to 95% for both backends and reduces the accumulated volume error to 2 mL.

Fig. 7 shows that the mean combined preservation score increases from 2.0 to 4.0 for RLT and from 2.4 to 3.8 for DSRL over five independent trials. These results show that TacEnvelope consistently improves interaction quality across different RL backends.

TABLE I

SHORT-HORIZON TACTILE FORCE PREDICTION AND UNCERTAINTY-SCALED ENVELOPE CALIBRATION ON STABLE-CONTACT TEST WINDOWS.

Task	Mean Force (N)	MAE (N) ↓	90% Cov. (%)
USB Insertion	51.02	1.32	89.80
Ink Dispensing	20.19	0.80	88.79
Pastry Grasping	3.48	0.60	91.25

TABLE II

ONLINE RL POST-TRAINING RESULTS FOR USB INSERTION AND INK DISPENSING.

Task	Method	Success (%) ↑	Quality ↓
USB Insertion	$\pi_{0.5}$	60	16 s
	RLT	72.5	19 s
	RLT + TacEnvelope	**85**	**12 s**
	DSRL	87.5	20 s
	DSRL + TacEnvelope	**95**	**13 s**
Ink Dispensing	$\pi_{0.5}$	60	5 mL
	RLT	90	3 mL
	RLT + TacEnvelope	**95**	**2 mL**
	DSRL	85	5 mL
	DSRL + TacEnvelope	**95**	**2 mL**

For USB insertion, Quality denotes mean completion time. For ink dispensing, it denotes accumulated absolute volume error over 10 dispensing operations. Lower values indicate better execution quality.

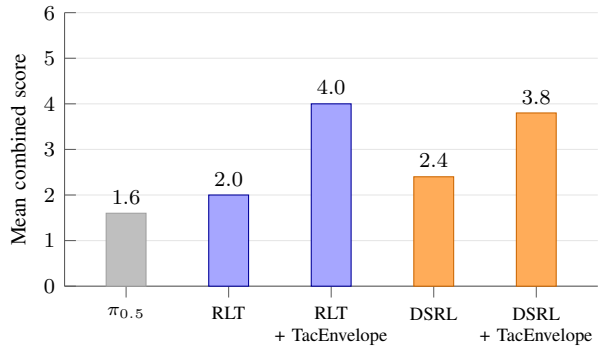


Fig. 7. Mean combined pastry preservation score after 20 consecutive grasp-and-transfer cycles over five independent trials. The maximum score is 6. Higher is better.

D. Ablation Study

1) Effect of Adaptive Force Envelopes: We compare adaptive force envelopes with a global fixed-band baseline across all three tasks. For each task, both methods use identical bilateral stable-contact windows from the same episode-level split.

For each task, the global fixed band is defined using the mean training contact load μ_{train} and the 5th and 95th percentiles, q_{05} and q_{95}, of the calibration residuals $r_i = y_i - \mu_{\text{train}}$, where y_i denotes the ground-truth contact load. The resulting interval is $[\mu_{\text{train}} + q_{05}, \mu_{\text{train}} + q_{95}]$.

Fig. 8 compares coverage (the fraction of observed loads within the bounds), mean interval width, and interval score ($\alpha = 0.1$), which penalizes width and out-of-interval deviations. Across all three tasks, adaptive envelopes achieve coverage closer to the nominal 90% target while reducing mean width by 34.3–91.8% and interval score by 59.9–85.7% relative to the global fixed band. These results support more precise characterization of interaction-dependent contact load in successful demonstrations.

2) Effect of Dual-Rate Inference: We evaluate dual-rate inference through paired offline comparisons on test-split episodes across all three tasks. Each 600 ms execution cycle contains two 300 ms tactile windows. Both settings are evaluated against the same measured second-window bilateral contact load, using identical expert models and calibration coefficients obtained only from the calibration split. Coverage is computed on a common subset of second-window samples in which bilateral contact has been established.

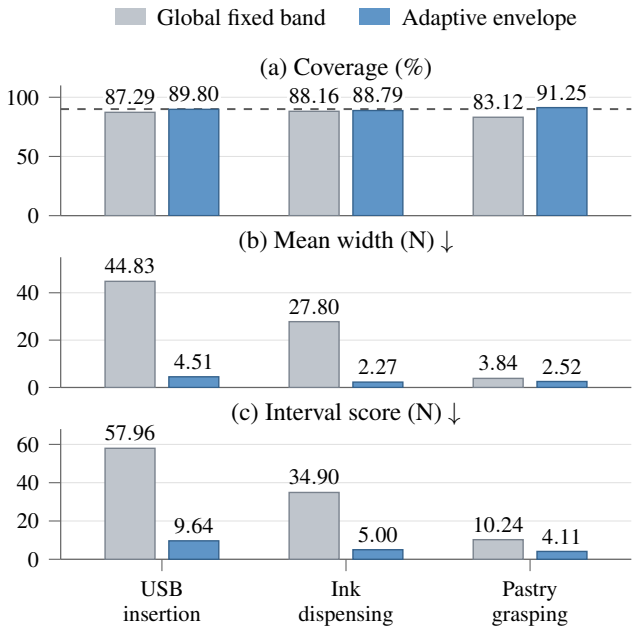


Fig. 8. Global fixed band versus adaptive envelope across three tasks. The dashed line marks the nominal 90% coverage target. Mean width and interval score ($\alpha = 0.1$) are reported in N, with lower values preferred.

TABLE III

SINGLE-RATE VERSUS DUAL-RATE TACTILE EVALUATION

Task	Rate Setting	Coverage (%)	$\text{IS}_{0.1}$ (N) ↓
USB Insertion	Single-rate	69.82	68.791
	Dual-rate	88.07	**8.893**
Ink Dispensing	Single-rate	73.84	30.424
	Dual-rate	89.74	**5.467**
Pastry Grasping	Single-rate	88.41	5.715
	Dual-rate	91.30	**4.144**

Paired offline evaluation on test-split episodes. Dual-rate keeps the initial z_{rl} fixed within each execution cycle, matching online execution. Coverage is computed on the same second-window subset with established bilateral contact. $\text{IS}_{0.1}$ denotes the interval score for nominal 90% prediction intervals; lower is better.

As shown in Table III, dual-rate inference brings second-window coverage closer to the nominal 90% level and lowers interval score across all three tasks. These results support the benefit of within-cycle envelope refreshing for tracking changes in contact load.

V. CONCLUSIONS

We introduced TacEnvelope, a framework for RL post-training of Vision-Language-Action policies. TacEnvelope learns a contact load envelope from successful demonstrations and converts tactile deviations into reward for policy optimization without modifying VLA policy. Real world experiments on USB insertion, ink dispensing, and pastry

grasping demonstrate that TacEnvelope improves physical interaction quality across diverse contact-rich manipulation tasks.

Future work will extend the tactile envelope from force magnitude to richer wrench information, including force direction and torque, for more comprehensive physical interaction tasks.

APPENDIX

A. Implementation Details

a) Data partitioning and temporal alignment.: For each task, episodes are partitioned into training, validation, calibration, and test sets with proportions of 70/10/10/10 using split seed 44. Normalization statistics are computed from the training episodes only. The validation split is used for checkpoint selection and early stopping, while the calibration split is used only to estimate the envelope coefficients.

Robot actions are represented at 30 Hz. The tactile window uses $H_T = 9$ control steps, corresponding to $T_w = 0.3\,\mathrm{s}$. Since the two tactile streams are asynchronous, tactile measurements are assigned to each window according to their timestamps, following Sec. III-A. Each tactile side must contain at least four valid measurements within a window. Missing tactile samples are not interpolated or replaced by assuming a fixed sensor frame rate.

b) Tactile preprocessing and causal history.: An episode-level force offset is estimated from the initial one-second contact-free interval and subtracted from each three-axis force measurement before computing its magnitude. A three-sample median filter is applied before constructing the Top2Mean prediction target. Unfiltered, bias-corrected force magnitudes are retained separately for peak-force monitoring.

The causal tactile history contains up to six valid samples per side from the preceding 200 ms. Each history sample contains the corrected three-axis force, its magnitude, and its relative timestamp. Missing history entries are masked rather than interpolated.

c) Envelope expert architecture and optimization.: In our $\pi_{0.5}$ implementation, the generic VLA representation z_t in Sec. III-A is instantiated as a frozen 512-dimensional RL-token representation z_{rl}. It is projected to dimension 256 and used as the VLA condition for the envelope expert.

The robot state is encoded as one token, while the nine actions within the tactile prediction horizon are encoded separately with learned positional embeddings. The bilateral tactile histories contribute up to twelve additional tokens. Two learned envelope queries are used to predict the left- and right-side Gaussian parameters.

The expert contains two Transformer decoder layers with hidden dimension 256, eight attention heads, feed-forward dimension 1024, and dropout 0.1. We train it using AdamW with learning rate 3×10^{-4}, weight decay 10^{-4}, batch size 128, and gradient clipping at norm 1. Training runs for at most 80 epochs, with early stopping based on the validation objective.

d) Online contact and reward implementation.: The main text defines the tactile penalty for windows with established contact. In the implementation, contact activation is determined causally from recent tactile measurements and is not inferred from future target forces. The resulting gate only determines whether the soft envelope penalty is active; peak-force monitoring is evaluated independently.

e) Dual-rate execution details.: The VLA predicts $H_a = 50$ actions, while each RL transition executes $H_e = 18$ control steps. Two tactile evaluations are therefore performed per transition using $H_T = 9$-step windows. Within the same execution cycle, the VLA representation is reused, while robot state, tactile history, action slice, and contact state are refreshed for the second tactile evaluation. The two valid tactile costs are averaged before being incorporated into the transition reward.

B. Additional Qualitative Results

Fig. 9 provides representative qualitative comparisons after 5, 10, 15, and 20 repeated grasp-and-transfer cycles. The TacEnvelope variants exhibit less visible progressive damage for both RLT and DSRL in these examples, consistent with the quantitative preservation scores reported in the main text. The qualitative results are intended to illustrate the temporal accumulation of physical damage under repeated contact rather than to provide an independent statistical evaluation.

C. Additional Ablation Results

We further analyze the key design choices of TacEnvelope, including action conditioning, probabilistic envelope learning, adaptive interval construction, and dual-rate inference. The USB expert ablations use training seeds 42, 43, and 44 with the same episode split and cached VLA features. Results are reported as mean $\pm$ population standard deviation across training seeds. Cross-task interval comparisons use seed 44.

a) Expert design.: We first evaluate the contribution of action conditioning and the probabilistic training objective on USB insertion. In the action ablation, the action projection, learned positional embeddings, and all nine action tokens are removed rather than replaced with zeros, while the remaining expert architecture and optimization settings are kept unchanged. Removing action conditioning increases MAE from 1.258 to $1.311\,\mathrm{N}$ and RMSE from 3.276 to $3.489\,\mathrm{N}$, indicating that demonstrated future actions provide useful information for short-horizon contact-load prediction.

We also replace the Gaussian NLL plus $0.5\,SmoothL1$ objective with SmoothL1 alone. As shown in Table IV, the probabilistic model achieves lower point-prediction error. The SmoothL1-only model uses additive residual calibration, whereas the probabilistic model uses uncertainty-scaled calibration; therefore, their coverage and width values compare the complete envelope designs rather than the training losses alone.

b) Adaptive envelope construction.: We next compare the adaptive envelope with a global fixed force band without retraining the expert. The fixed baseline uses one task-level interval whose center and width remain constant across all

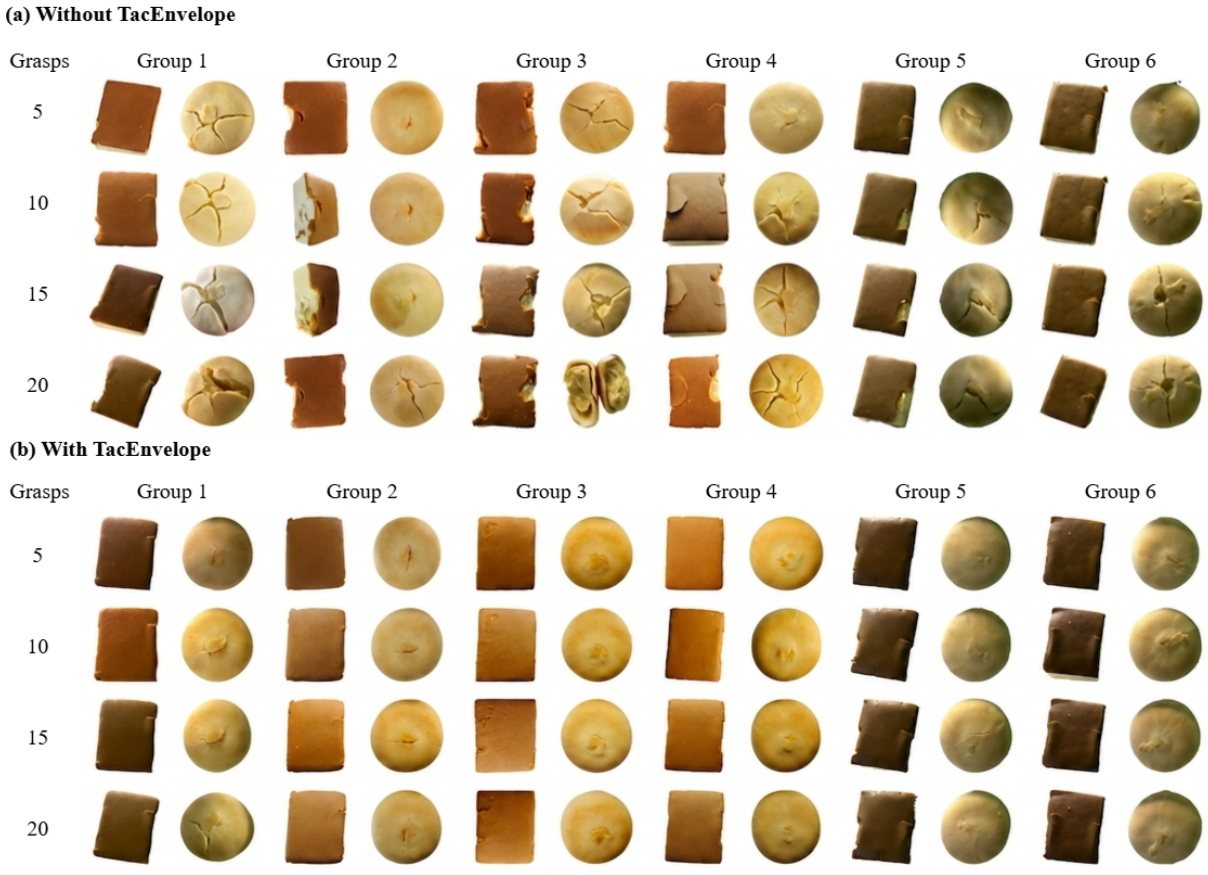


Fig. 9. Representative pastry damage after 5, 10, 15, and 20 repeated grasp-and-transfer cycles using task-adapted $\pi_{0.5}$, RLT, DSRL, and their TacEnvelope variants. Each group contains cake and mung bean cake.

TABLE IV

USB EXPERT ABLATIONS ON IDENTICAL BILATERAL STABLE-CONTACT WINDOWS. VALUES ARE MEAN $\pm$ STANDARD DEVIATION OVER THREE TRAINING SEEDS. COVERAGE HAS A NOMINAL TARGET OF 90%.

Expert	Calibration	MAE (N)	RMSE (N)	Cov. (%)	Width (N)
Full probabilistic	Scaled	1.258 ± 0.049	3.276 ± 0.097	90.02 ± 0.32	4.104 ± 0.294
w/o action	Scaled	1.311 ± 0.104	3.489 ± 0.176	90.75 ± 1.10	4.058 ± 0.054
SmoothL1 only	Additive	1.405 ± 0.087	3.665 ± 0.377	89.74 ± 1.43	4.357 ± 0.203

interaction states, whereas the adaptive envelope changes both according to the current VLA and interaction context.

Table V shows that the adaptive envelope achieves coverage closer to the nominal 90% target while substantially reducing interval width and interval score across all three tasks. On USB insertion, for example, the mean width decreases from 44.831 N to 4.513 N while coverage increases from 87.29% to 89.80%. This suggests that a single global force range is too coarse to characterize the contact-load variation observed during successful manipulation.

We additionally examine three bilateral uncertainty aggregation rules on the USB seed-44 model. Mean, root-sum-square (RSS), and maximum aggregation produce coverage/width pairs of $89.80\%/4.513$ N, $90.13\%/4.529$ N, and $88.29\%/4.460$ N, respectively. Mean and RSS behave similarly, while maximum aggregation gives slightly lower coverage.

TABLE V

GLOBAL FIXED BAND VERSUS ADAPTIVE ENVELOPE USING SEED 44. PARENTHESES INDICATE THE NUMBER OF TEST WINDOWS.

Task	Method	Cov. (%)	Width (N)	$IS_{0.1}$ (N)
USB (598)	Fixed	87.29	44.83	57.96
	Adaptive	89.80	4.51	9.64
Ink (642)	Fixed	88.16	27.80	34.90
	Adaptive	88.79	2.27	5.00
Pastry (160)	Fixed	83.12	3.84	10.24
	Adaptive	91.25	2.52	4.11

c) Dual-rate inference.: We finally evaluate the dual-rate scheme using paired offline counterfactuals on recorded test trajectories. Both settings retain the initial VLA representation z_{rl} throughout an $H_e = 18$ execution cycle. The single-rate variant retains the first-window envelope and contact state for the second tactile window, whereas dual-

TABLE VI

PAIRED SECOND-WINDOW EVALUATION WITH FIXED INITIAL z_{rl}. PARENTHESES INDICATE ELIGIBLE TARGETS.

Task	Setting	Cov. (%)	$\mathrm{IS}_{0.1}$ (N)
USB (285)	Single	69.82	68.791
	Dual	88.07	8.893
Ink (302)	Single	73.84	30.424
	Dual	89.74	5.467
Pastry (69)	Single	88.41	5.715
	Dual	91.30	4.144

rate refreshes robot state, causal tactile history, the next $H_T = 9$ scheduled actions, and contact state. Both settings are evaluated against the same measured second-window contact load.

As shown in Table VI, dual-rate inference moves coverage substantially closer to the nominal 90% target for USB and ink while strongly reducing interval score. The improvement is smaller for pastry, where the single-rate baseline already achieves 88.41% coverage. These results support more frequent envelope updates for tracking within-cycle contact changes, but do not constitute a separate online-RL success comparison.

As an additional diagnostic, we compare material second-window soft-penalty events between the held and refreshed computations. Relative to the dual-rate computation, the single-rate variant misses/adds $4/37$, $5/48$, and $2/10$ events for USB, ink, and pastry, respectively. These values characterize disagreement between the two reward computations rather than false-positive or false-negative rates against an external ground-truth reward.

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